

CoRoT-2b

R-1756

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_130707 · created 2026-08-02T20:26:12+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456480.7736 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.223 +/- 0.065
Transit depth (Rp/Rs)^2	4.96 +/- 2.87 [%]
Orbital Inclination (inc)	83.34 +/- 2.08
Airmass coefficient 1 (a1)	11.04 +/- 8.96
Airmass coefficient 2 (a2)	-1.62 +/- 0.77
Scatter in the residuals of the lightcurve fit is	103.67 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	5.75
Transit Duration (day)	0.073 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.223 ± 0.065 vs 0.1667 (deviation 0.87σ)
Duration fit vs published	0.073 d vs 0.095 d (deviation 1.05σ)
Mid-time O–C	-0.0 min
Depth SNR	3.4
Residual scatter	103.67 %

Light curve

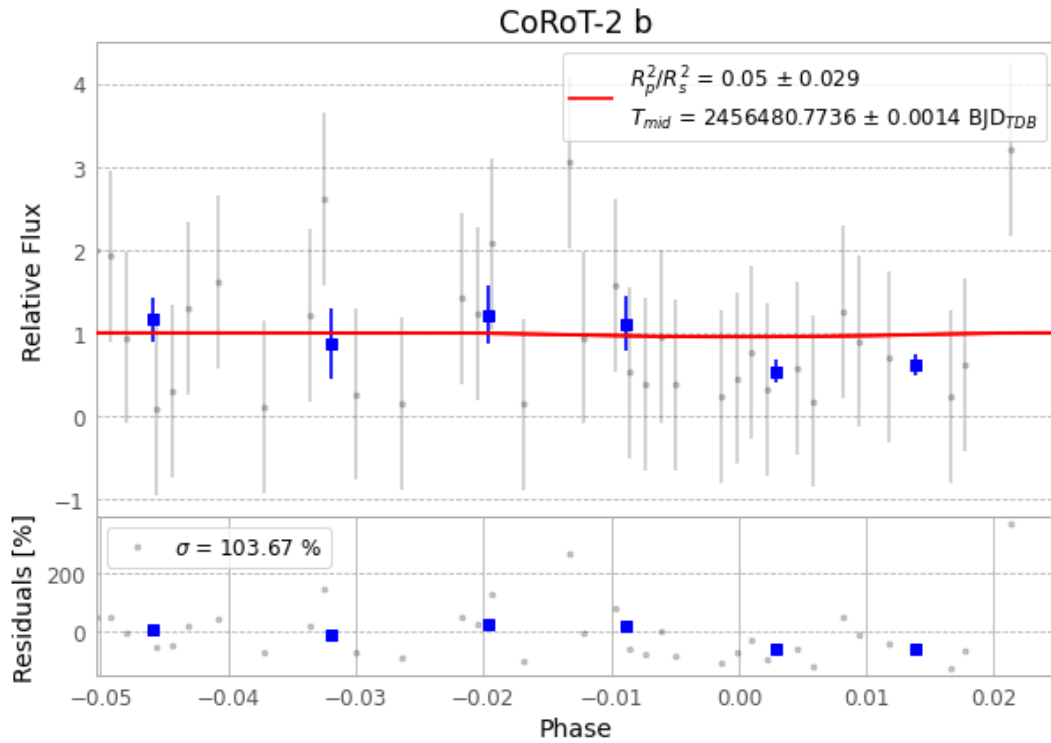


Figure 1 — FinalLightCurve_CoRoT-2 b_2013-07-06.png (SHA-256 59777373d9199d8b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [308, 185], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 7667ce7f075eed93...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), CoRoT-2130707041812.FITS → CoRoT-2130707072712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-130707.log (5cdd0f41e47a...), FinalLightCurve_CoRoT-2 b_2013-07-06.png (59777373d919...), FinalLightCurve_CoRoT-2 b_2013-07-06.csv (ca2ace611dc6...)

Compute resources

Wall time 190.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 20:26Z.